

# MAT 115E Introduction to Programming Language

Lab-6/ CRN : 23492

---

Instructor: Dr. Evren Tanrıöver

Lab Assistant: Res. Asst. Tugce Sezer

---

## 1 Question 1

Each digit in a non-negative integer  $k$  has a digit count. For example, in 668889, the digit 6 appears 2 times and the digit 8 appears 3 times, while the digit 9 appears only once. The digits 0, 1, 2, 3, 4, 5, and 7 do not appear at all.

Write a C function named **digitCount** that takes a non-negative integer  $k$  and a digit  $d$ , and returns the number of times  $d$  appears in  $k$ .

## 2 Question 2

The explicit form of Maclaurin series expansion for  $e^x$  is given below.

$$e^x = \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{x^k}{k!} = x^0 + \frac{x^1}{1!} + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \cdots \quad (1)$$

Let  $a_n$  is the  $n$ -term in the series, then the sum is given as following recursion formula:

$$S_0 = a_0$$

$$S_n = S_{n-1} + a_n \quad \text{for } n \geq 1$$

.

Write a **C function** that takes a positive integer number  $n$  and a real number  $x$  as parameters and returns the first  $n$ -term summation of the Maclaurin series expansion for  $e^x$ .

## 3 Question 3

Consider an ISBN (International Standart Book Number) with 11 digits. The last digit of this number is used as a check digit. Whether an ISBN is valid or not can be controlled as follows. Let  $x_1, x_2, \dots, x_{11}$  be the digits of ISBN from the left to the right.

$$(10 * x_1 + 9 * x_2 + 8 * x_3 + 7 * x_4 + 6 * x_5 + 5 * x_6 + 4 * x_7 + 3 * x_8 + 2 * x_9 + x_{10}) \bmod 10 = x_{11}$$

Write a C program that reads an ISBN number and checks whether it is valid or not.

Example Scenario:

Assume that user enters 4720611384**3**

$$(10 * 4 + 9 * 7 + 8 * 2 + 7 * 0 + 6 * 6 + 5 * 1 + 4 * 1 + 3 * 3 + 2 * 8 + 1 * 4) \bmod 10 = 3$$

Since match the last digit, ISBN is valid. Therefore, your function returns 1.